14

Introduction to the Algebraic Formulation of Quantum Theories

I would like to make a confession which may seem immoral: I do not believe absolutely in Hilbert spaces any more.
von Neumann, letter to Birkhoff about the mathematical formulation of QM (1935)

In the last chapter of the book we offer a short presentation of the algebraic formulation of quantum theories, and we will state and prove a central theorem about the so-called *GNS construction*. We will discuss how to treat the notion of quantum symmetry in this framework, by showing that an algebraic quantum symmetry can be implemented (anti-)unitarily in GNS representations of states invariant under the symmetry.
As general references, mostly concerned with the algebraic formulation of Quantum Field Theories, we recall [Emc72], [Haa96], [Ara09], [Rob04], [BDFY15] and the more recent [Str05a, Str12, Lan17] on the algebraic formulation of QM. On the mathematical side, detailed and critical studies on the present material are [BrRo02] and [KaRi97].
We will routinely resort to definition 3.52 throughout the chapter.

14.1 Introduction to the algebraic formulation of Quantum Theories

The fundamental theorem 11.43 of Stone-von Neumann is stated in the jargon of theoretical physics as follows:
"*all irreducible representations of the CCRs with a finite, and fixed, number of freedom degrees are unitarily equivalent*,".
The expression *unitarily equivalent* refers to the existence of a Hilbert-space isomorphism S, and the finite number of degrees of freedom is the dimension of the symplectic space X on which the Weyl algebra is built.
What happens then in infinite dimensions? Let us keep irreducibility, and suppose we pass from X finite-dimensional – parametrising, e.g., the coordinates of a point-particle in phase space – to X infinite-dimensional – describing a suitable solution space to free bosonic field equations, say. Then the Stone–von Neumann theorem no longer holds. Theoretical physicists would say that
"*there exist non-equivalent irreducible CCR representations with an infinite number of freedom degrees*".

What happens in this situation, in practice, is that one can find strongly continuous irreducible representations π_1, π_2, on (separable) Hilbert spaces H_1, H_2, of the Weyl *-algebra $\mathfrak{A} := \mathscr{W}(\mathsf{X},\sigma)$ (here thought of as C^*-algebra, with no change in the results) associated to the physical system under exam (a quantised bosonic field, typically), that admit *no* isomorphism $S : \mathsf{H}_1 \to \mathsf{H}_2$ satisfying:

$$S\pi_1(a)\,S^{-1} = \pi_2(a)\,, \quad \text{for any } a \in \mathfrak{A}.$$

Pairs of this kind are called *(unitarily) non-equivalent*. Jumping from X finite-dimensional to infinite-dimensional corresponds to passing from Quantum Mechanics to Quantum Field Theory (relativistic QFT, possibly, and on curved spacetime [Wal94, KhMo15, FeVe15]). In these situations (but not only), the existence of non-equivalent representations has often to do with *spontaneous symmetry breaking*. The presence of non-equivalent representations of a single physical system (X,σ) shows that a formulation in a fixed Hilbert space is completely inadequate, and we must free ourselves of the Hilbert structure in order to lay the foundations of quantum theories in broader generality (an interesting and detailed technical analysis of several problems with the Hilbert-space formulation, based on concrete models of quantum fields and statistical mechanics' systems, can be found in [Emc72, Sect.1, Ch.1]).
This programme has been developed by and large, starting from the pioneering work of von Neumann himself, and is nowadays called *algebraic formulation of Quantum (Field) Theories*. Within this framework it was possible to formalise, for example, field theories on curved spacetime in relationship to the quantum phenomenology of black-hole thermodynamics.

14.1.1 Algebraic formulation

The algebraic formulation prescinds, anyway, from the nature of the quantum system, and may be stated for systems with finitely many freedom degrees as well [Str05a]. The viewpoint falls back on two assumptions [Haa96, Ara09, Str05a, Str12, BDFY15] (which somehow generalise the results of §7.6.5).

AA1. *A physical system S is described by its* **observables***, viewed now as self-adjoint elements in a certain C^*-algebra $\mathfrak{A}_S$ with unit $\mathbb{I}$ associated to S.*

AA2. *An* **algebraic state** *on $\mathfrak{A}_S$ is a linear functional $\omega : \mathfrak{A}_S \to \mathbb{C}$ such that:*

$$\omega(a^*a) \geq 0 \quad \forall\, a \in \mathfrak{A}_S, \qquad \omega(\mathbb{I}) = 1\,,$$

that is, positive *and* normalised to 1.

We have to stress that $\mathfrak{A}_S$ is not seen as a concrete C^*-algebra of operators on a given Hilbert space, but remains an abstract C^*-algebra. Physically, $\omega(a)$ is the *expectation value* of the observable $a \in \mathfrak{A}$ in state ω.

Remark 14.1.
(1) $\mathfrak{A}_S$ is usually called *the algebra of observables of S* though, properly speaking, the observables are the self-adjoint elements of $\mathfrak{A}_S$ only.
(2) We can assign $\mathfrak{A}_S$ to S irrespective of any reference frame, provided we assume that the (active) transformations of the various frames are given by automorphisms of $\mathfrak{A}_S$. If the algebra depends on the frame, the time evolution *with respect to the given frame* is described by a one-parameter group of *-automorphisms $\{\alpha_t\}_{t\in\mathbb{R}}$, where $\alpha_t : \mathfrak{A}_S \to \mathfrak{A}_S$ is a *-homomorphism for any $t \in \mathbb{R}$, α_0 is the identity and $\alpha_t \circ \alpha_{t'} = \alpha_{t+t'}$ for $t,t' \in \mathbb{R}$. It is natural to demand weak continuity in t: for every state ω on $\mathfrak{A}_s$, the map $\mathbb{R} \ni t \mapsto \omega(\alpha_t(a))$ is continuous for every $a \in \mathfrak{A}_S$.
In case the algebra of observables is independent of the reference frame, $\mathfrak{A}_S$ is actually thought of as a *net of algebras*: it will be described by a function $\mathscr{O} \mapsto \mathfrak{A}_S(\mathscr{O})$ mapping to an algebra $\mathfrak{A}_S(\mathscr{O})$ any regular bounded region $\mathscr{O}$ in spacetime (extending both in the space and time directions). From this point of view time evolution is replaced by causal relations between algebras localised at spacetime regions that are causally related, in particular when one region belongs to another's future. The above approach, thoroughly discussed for the first time in the crucial paper of Haag and Kastler [HaKa64], is the modern stepping stone to develop algebraic field theory in the local covariant formulation.
(3) It is important to emphasise that, differently form the Hilbert space formulation, the algebraic approach can be adopted to describe *both classical and quantum systems*. The two cases are distinguished on the base of the commutativity of the algebra of observables $\mathfrak{A}_S$. A commutative algebra is assumed to describe a classical system, whereas a noncommutative one is supposed to be associated with a quantum system.
■

14.1.2 Motivations and relevance of Lie-Jordan algebras

Before going on with the mathematical technology, a brief discussion about the motivations underlying the algebraic formulation may be useful. In the rest of this subsection, whose content is quite heuristic, we shall denote the observables with capital letters $A,B,\dots$ and use corresponding small letters $a,b,\dots$ to denote the values attained by the observables.
The most evident, *a posteriori*, justification of the algebraic approach lies in its powerfulness [Haa96]. There have been a host of attempts to account for assumptions **AA1** and **AA2** and their physical meaning, in full generality (see the study of [Emc72], [Ara09] and [Str05a, Str12] and especially the work of I. E. Segal [Seg47] based on *Jordan algebras*), yet none seems to be definitive [Stre07].
There are at least two interrelated basic issues to consider when attempting to justify assumptions **AA1** and **AA2** physically.

(a) *The identification of the notions of state and expectation value.*

This identification would be natural within the Hilbert space formulation, where the class of observables includes elementary observables, which are represented by or-

thogonal projectors and correspond to "yes-no" statements. The expectation value of one such observable coincides with the probability that the outcome of the measurement is "yes". The set of all those probabilities defines, in fact, a quantum state of the system, as we know. The analogues of these elementary propositions, however, generally do not belong to the C^*-algebra of observables in the algebraic formulation (see §14.1.6). Even so, this obstruction is not insurmountable. Following [Ara09], in a completely general physical system the most general notion of state ω would be the assignment of all probabilities $w_\omega^{(A)}(a)$ that the outcome of measuring the observable A is a, for all observables A and all values a. On the other hand, it is known [Str05a] that all experimental information on the measurement of A in state ω – the probabilities $w_\omega^{(A)}(a)$ in particular – is recorded in the expectation values of the polynomials of A. Here, we should think of $p(A)$ as the observable whose values are given by the $p(a)$, for all values a of A. This characterisation of an observable is theoretically supported by the various solutions to the *momentum problem* in probability theory. To adopt this paradigm, therefore, we have to assume that the set of observables must include all real polynomials $p(A)$ whenever it contains the observable A. This is in agreement with the much stronger requirement **AA1**.
We stress that, within the algebraic formulation, it is natural to suppose that the full set of observables and the whole set of states together encompass the maximum amount of information available on the physical system we are studying. As a byproduct of this assumption and of the identification of the notion of state with that of expectation value, we are committed to conclude that, in our formalism:

(i) **states separate observables**: two observables A, B coincide if and only the corresponding expectation values coincide: $A = B \Leftrightarrow \omega(A) = \omega(B)$ for all states ω;

(ii) **observables separate states**: two states ω, ω' coincide if and only if the corresponding expectation values coincide: $\omega = \omega' \Leftrightarrow \omega(A) = \omega'(A)$ for all observables A.

As a matter of fact, **AA1** and **AA2** theoretically support these two statements. Indeed, (ii) follows immediately from the fact that a state is a linear functional on the C^*-algebra $\mathfrak{A}_S$ and that every element (also not self-adjoint) of $\mathfrak{A}_S$ is a linear combination of self-adjoint elements representing observables. Assertion (i) is a straightforward consequence of the Gelfand-Najmark theorem, as we shall see in corollary 14.30.

(b) *The incarnation of observables as a C^*-algebra.*

Whilst the nature and the physical meaning of the associative product of the algebra, in particular, is difficult to justify *a priori*, the assumption on the boundedness of observables in a suitable norm does not seem so hard to clarify from an operational point of view [Str05a]. An observable A is defined in terms of a concrete experimental apparatus, which yields the numerical results of measurements in any state ω. Since each concrete experimental apparatus has inevitable limitations that imply a scale bound independent of the state on which measurements are performed, the result of measurements of A on the various states is a bounded set of real numbers, whose bound is related to the scale bound of the associated experimental apparatus. It is then natural to associate to each observable A a finite bound (in perfect agreement with the theoretical result of corollary 14.30):

$$||A|| := \sup_{\omega \in \Omega} |\omega(A)| < +\infty \tag{14.1}$$

where Ω is the set of all possible states. As the notation suggests, this bound can be interpreted as a norm, once we have built the algebraic structure of the observables (see [Str05a] for details). Let us next focus on the purely algebraic features of the space of observables, the associative algebra of product in particular. In this respect we can observe that, actually, a weaker form of **AA1** must certainly be true. As already noticed, if p is a real polynomial and A is an observable (so that the possible values a are real numbers), $p(A)$ is a well-defined object and it is an observable as well. As we said $p(A)$ is nothing but the observable whose values are $p(a)$, for all values a of A. If p is complex, $p(A)$ deserves the same operative interpretation since its real and imaginary parts are observables. All this indicates the existence, for every observable A, of a natural structure of a commutative *-algebra on complex polynomials $p(A)$. The involution here is the obvious one $p(A)^* := \overline{p}(A)$. It is not difficult to prove that (14.1) turns out to be a norm for this *-algebra.
Following the same route of §11.3.2, objects like real linear combinations $\alpha A + \beta B$ of observables can be made physically meaningful even if A and B cannot measured simultaneously: $\alpha A + \beta B$ is the observable whose expectation values verify $\omega(\alpha A + \beta B) = \alpha\omega(A) + \beta\omega(B)$ for every state ω. Notice that, from the physical point of view, we are enlarging the class of observables, for we are assuming that new observables verifying the previous constraint exist (and thus they are uniquely determined, since their expectation values are known). The extension to complex combinations is now straightforward. All that on the one hand justifies the appearance of a complex vector space structure, equipped with an antilinear bijective and involutive operation (extending the previous one), $(\alpha A + \beta B)^* := \overline{\alpha}A + \overline{\beta}B$, and with a norm that extends (14.1), as is not difficult to prove. This structure becomes a fully fledged normed *-algebra if the construction is supplemented by an associative product, that extends the one defined in each algebra of polynomials $p(A)$ generated by any fixed element A. On the other hand, the procedure gives rise to a generally *non-distributive* and *non-associative Jordan product* (see §11.3.2 and definition 11.29):

$$A \circ B := \frac{1}{2}\left((A+B)^2 - A^2 - B^2\right) , \tag{14.2}$$

where $(A+B)^2$, A^2 and B^2 are well defined as indicated above.
However, similarly to what happens in the Hilbert space formulation, if p denotes a general polynomial with two entries, and A and B are observables that *cannot be measured simultaneously* (see §11.3.2), the interpretation of $p(A,B)$ turns out to be very difficult. Certainly, the product of the C^*-algebra of **AA1** provides a sound *mathematical* meaning to $p(A,B)$, thought it does not elucidate the *physical* meaning of $p(A,B)$ in terms of A and B. That said, it is difficult to justify the existence of a

C^*-algebra product in the complex vector space of extended observables, even assuming the existence of a meaningful Jordan product and norm [Str05a]. Necessary and sufficient conditions have been found (e.g., see [AlSc03]) for the existence of a C^*-algebra structure that completes the algebraic and topological structures outlined above. However the physical interpretation of these conditions does not seem transparent. (Sect.2, Ch.1 of [Emc72] contains a deep study on the Jordan-algebra induced C^*-structure, and the many steps involved are analysed thoroughly.)
Another, much stronger, possibility is to assume – without any true physical justification – that the real linear space of observables is also equipped with a Lie-algebra structure of pure quantum nature (the Lie bracket $[A,B]$ is proportional to the Planck constant), and that the two products enjoy the properties below.

Definition 14.2. *A (real) Jordan algebra* $\mathfrak{A}$ *(definition 11.29) equipped with a Lie bracket* $\{\ ,\ \}$ *is called (real)* **Lie-Jordan algebra** *when, for a constant* $c \in \mathbb{R}$*, it satisfies:*

$$(A \circ B) \circ C - A \circ (B \circ C) = \frac{c}{4}\{\{A,C\},B\} \quad \textit{for all } A,B,C \in \mathfrak{A} \tag{14.3}$$

and furthermore the **Leibniz rule** *is valid:*

$$\{A, B \circ C\} = \{A,B\} \circ C + \{A,C\} \circ B \quad \textit{for all } A,B,C \in \mathfrak{A}. \tag{14.4}$$

From the physical side, in our context it is natural to assume $c = \hbar^2$. Moreover, if we look at the Hamiltonian formulations of classical physics for instance, $\{\ ,\ \}$ could be interpreted as the quantum analogue of the Poisson bracket. In other words the Lie bracket should be understood as the standard commutator of (self-adjoint) observables multiplied by $i\hbar$, in accordance with a version of Dirac's correspondence principle (see §11.5.8).
A posteriori, the Lie-Jordan structure arises quite naturally when one knows that (complexified) observables form a *-algebra. In fact, given a complex *-algebra, by defining the Lie bracket in terms of the standard commutator as:

$$\{\ ,\ \} := \frac{i}{\hbar}[\ ,\ ] \tag{14.5}$$

and setting

$$A \circ B = \frac{1}{2}(AB + BA)\ , \tag{14.6}$$

one finds a natural real Lie-Jordan algebra when restricting to the real space of self-adjoint elements. In particular:

$$AB = A \circ B + \frac{i\hbar}{2}\{A,B\} \quad \text{for all } A,B,C \in \mathfrak{A}. \tag{14.7}$$

Taking (14.6) into account, that is the same as:

$$AB = A \circ B + \frac{1}{2}[A,B] \quad \text{for all } A,B,C \in \mathfrak{A}\,. \tag{14.8}$$

The procedure we have outlined can be reversed if we assume that observables possess a Lie-Jordan structure. Given a real Lie-Jordan algebra $\mathfrak{A}$ with $c = \hbar^2$, requiring (14.5) produces an associative complex *-algebra. We can perform this on $\mathfrak{A}_1 := \mathbb{C} \otimes \mathfrak{A}$: the involution * is $(a \otimes A)^* := \overline{a} \otimes A$ for every $a \in \mathbb{C}$ and $A \in \mathfrak{A}$, and the associative product is given by extending (14.8) $\mathbb{C}$-linearly.
It is important to notice that the Lie-Jordan algebra structure can be equipped with a natural norm. After completion, this will generate an associated C^*-algebra [AlSc03]. Coming back to physics, it is finally worth stressing that with this formulation both the non-commutativity and non-associativity underlying the algebra of observables appear to have a quantum nature, and seem to realise quantum *deformations* of classical structures, in the sense that commutativity and associativity are restored as soon as $\hbar \to 0$. However, a relic of quantum non-commutativity remains in the Poisson bracket even at classical level. As a matter of fact, the picture we have described can be developed further. The (algebraic) *deformation quantisation* procedure, introduced in §11.5.8, starts by assuming that (14.7) is just the first-order expansion in $\hbar$ of the quantum product. This approach has been exploited in algebraic Quantum Field Theory in a modern and powerful fashion to reformulate the perturbative interacting theory, including the renormalisation procedure [BrFr09].

14.1.3 The GNS reconstruction theorem

The set of algebraic states on $\mathfrak{A}_S$ is a convex subset in the dual $\mathfrak{A}'_S$ of $\mathfrak{A}_S$: if ω_1 and ω_2 are positive and normalised linear functionals, $\omega = \lambda\omega_1 + (1-\lambda)\omega_2$ is clearly still positive and normalised, for any $\lambda \in [0,1]$.
Hence, just as we saw for the standard formulation, we can define *pure algebraic states* as extreme elements of the convex body.

Definition 14.3. *An algebraic state $\omega : \mathfrak{A} \to \mathbb{C}$ on the unital C^*-algebra $\mathfrak{A}$ is called a* **pure algebraic state** *if it is extreme in the set of algebraic states. An algebraic state that is not pure is called* **mixed**.

Later we will show that the space of states is non-empty and compact in the *-weak topology. Consequently, pure states exist.

Surprisingly, most of the abstract apparatus given by a C^*-algebra and a set of states admits elementary Hilbert space representations once a reference algebraic state has been fixed. This is by virtue of a famous procedure Gelfand, Najmark and Segal came up with, that we present below [Haa96, Ara09, Str05a].

Theorem 14.4 (GNS theorem). *Let $\mathfrak{A}$ be a C^*-algebra with unit $\mathbb{I}$ and $\omega : \mathfrak{A} \to \mathbb{C}$ a positive linear functional with $\omega(\mathbb{I}) = 1$. Then*
(a) *there exist a triple $(\mathsf{H}_\omega, \pi_\omega, \Psi_\omega)$, where H_ω is a Hilbert space, $\pi_\omega : \mathfrak{A} \to \mathfrak{B}(\mathsf{H}_\omega)$ a $\mathfrak{A}$-representation over H_ω and $\Psi_\omega \in \mathsf{H}_\omega$, such that:*
(i) Ψ_ω is cyclic for π_ω: the invariant subspace $\mathscr{D}_\omega := \pi_\omega(\mathfrak{A})\Psi_\omega$ is dense in H_ω;
(ii) $(\Psi_\omega|\pi_\omega(a)\Psi_\omega) = \omega(a)$ for every $a \in \mathfrak{A}$.
(b) *If (H, π, Ψ) satisfies (i) and (ii), there exists a unitary operator $U : \mathsf{H}_\omega \to \mathsf{H}$ such that $\Psi = U\Psi_\omega$ and $\pi(a) = U\pi_\omega(a)U^{-1}$ for any $a \in \mathfrak{A}$.*